sparsity_jac

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This file is part of CasADi.

CasADi -- A symbolic framework for dynamic optimization.
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1 sparsity_jac

[1]:
```
import casadi as ca
from numpy import *
import casadi as c
from pylab import spy, show
```

We construct a simple SX expression

[2]:
```
x = ca.SX.sym("x",40)
y = x[:-2]-2*x[1:-1]+x[2:]
```

Let's see what the first 5 entries of y look like

[3]:
```
print(y[:5])
```

```
[((x_0-(2.*x_1))+x_2), ((x_1-(2.*x_2))+x_3), ((x_2-(2.*x_3))+x_4),
((x_3-(2.*x_4))+x_5), ((x_4-(2.*x_5))+x_6)]
```

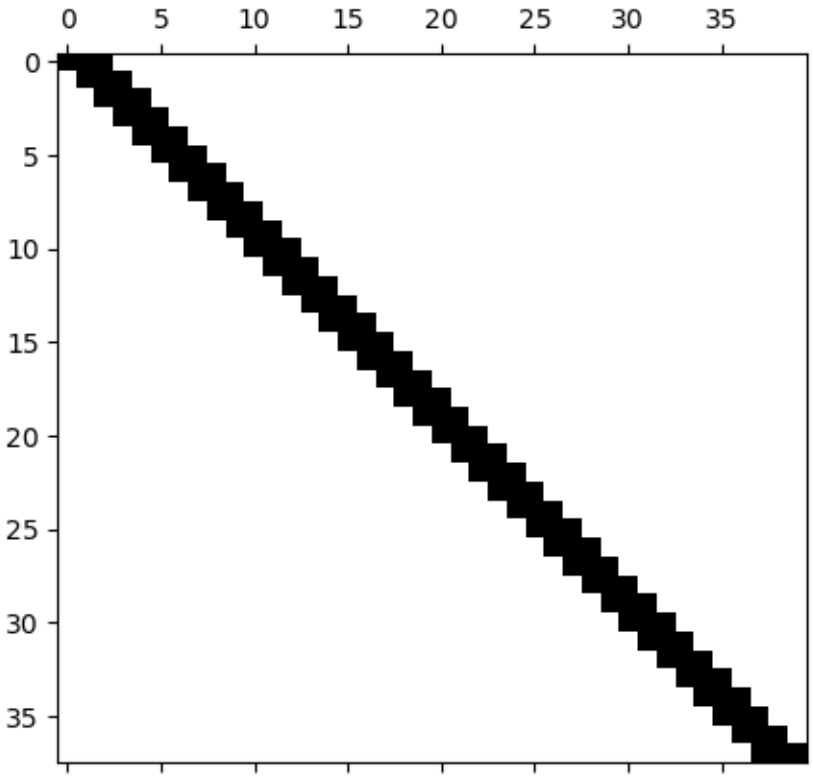
Next, we construct a function

[4]:
```
f = ca.Function("f", [x],[y])
```

And we visualize the sparsity of the jacobian

[5]:
```
spy(f.sparsity_jac(0, 0))
```

[5]:
```
<matplotlib.image.AxesImage at 0x7fdb40b588e0>
```



[6]:
```
show()
```